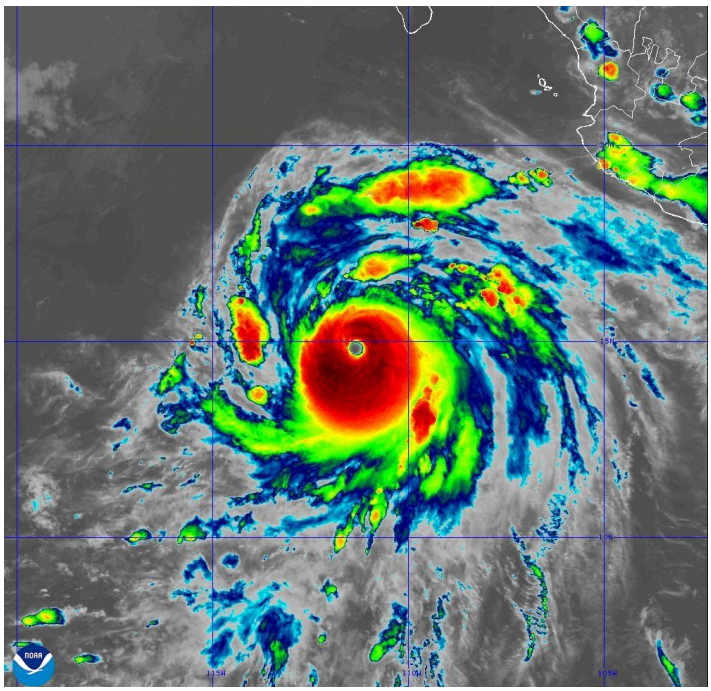
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE GENEVIEVE

(EP072026)

24 July–2 August 2026

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National Hurricane Center
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GOES-WEST GEOCOLOR IMAGE OF HURRICANE GENEVIEVE NEAR PEAK INTENSITY AT 0350 UTC 27 JULY 2026. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Genevieve was a Category 5 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that remained over the open waters of the eastern Pacific

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Genevieve.

Hurricane Genevieve

24 July–2 August 2026

BEST TRACK

The "best track[2]" positions and intensities for Hurricane Genevieve are listed in Table 1. The "best track" chart of Genevieve's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no ship observations of tropical-storm-force winds associated with Genevieve.

Origin

Genevieve developed from a tropical wave that moved off the west coast of Africa on 13 July. On 21 July, the tropical wave crossed Central America and entered the eastern Pacific basin. Favorable environmental conditions allowed the disturbance to gradually organize over the next few days. Deep convection became sufficiently organized around the low-level circulation center, leading to the formation of a tropical depression at 0600 UTC 24 July, about 500 n mi south of Acapulco, Mexico.

Peak Intensity and Minimum Pressure

Genevieve's peak intensity of 140 kt at 0600 UTC 27 July is based on a blend of subjective Dvorak and objective satellite intensity estimates. The objective estimates include an ADT value of T7.0/140 kt, UW-CIMSS D-MINT estimates of 138–142 kt, and SATCON values of 140–143 kt. Additionally, the subjective Dvorak intensity estimate from TAFB was T7.0/140 kt. Genevieve went through a period of rapid intensification from 1200 UTC 25 July through 0600 UTC 27 July, strengthening from a 50-kt tropical storm to a 140-kt category 5 hurricane over only 42 h.

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[2] A digital record of the complete best track, including wind radii, can be found on line at https://ftp.nhc.noaa.gov/atcf/. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Genevieve.

The estimated minimum central pressure of 924 mb is based on the Knaff-Zehr-Courtney (KZC) pressure-wind relationship. Pressure estimates based on the Dvorak pressure-wind relationship were a couple of millibars lower.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Genevieve.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Genevieve. The system that became Genevieve was first mentioned in the TWO 174 h prior to genesis with a low chance (<40%) of development within 7 days. These formation probabilities were increased to the medium category (40–60%) 126 h and the high category (>60%) 78 h before development. The 2-day formation probabilities were introduced 54 h before formation, and raised to the medium and high categories 36 h and 12 h before development, respectively. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Genevieve. Every outlook encompassed Genevieve's genesis location, and the tropical cyclone formed near the middle of the area denoted in most outlooks.

A verification of NHC official track forecasts for Genevieve is given in Table 3a. Official track forecast errors were significantly lower than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The official track forecasts (OFCL) outperformed the entire track guidance suite at the 60–120 h lead times. At 12–48 h, OFCL errors were comparable to the top-performing aids, with the consensus models TVCN and HCCA exhibiting the lowest errors during that period.

A verification of NHC official intensity forecasts for Genevieve is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. The official intensity forecasts (OFCL) generally performed well compared to the intensity guidance suite at most lead times. However, a few models had slightly lower errors than OFCL at various forecast hours. NHC intensity forecasts also exhibited exceptional skill during the cyclone's rapid intensification (RI) phase. Mean intensity errors were notably lower than the long-term averages typically associated with hurricanes that undergo R and reach category 5 intensity. The NHC consistently and correctly anticipated the onset of the RI episode well in advance of its occurrence. Although the official forecasts slightly underpredicted the system's peak intensity, the overall timing of the intensification was remarkably accurate (Fig. 5).

There were no coastal watches or warnings issued for Genevieve.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the Graphical TWO verification figure (Fig. 4) and the OFCL intensity figure (Fig. 5).

Table 1. Best track for Hurricane Genevieve, 24 July–2 August 2026.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
24 / 0600	8.6	99.1	1005	30	tropical depression
24 / 1200	8.7	100.2	1003	35	tropical storm
24 / 1800	8.9	101.2	1001	40	"
25 / 0000	9.3	102.1	1000	45	"
25 / 0600	9.9	103.0	1000	45	"
25 / 1200	10.5	103.9	997	50	"
25 / 1800	11.1	105.0	995	55	"
26 / 0000	11.8	106.2	991	65	hurricane
26 / 0600	12.5	107.4	980	80	"
26 / 1200	13.2	108.7	967	100	"
26 / 1800	13.8	109.8	950	115	"
27 / 0000	14.4	110.8	934	130	"
27 / 0600	15.0	111.7	924	140	"
27 / 1200	15.7	112.6	929	135	"
27 / 1800	16.4	113.5	934	130	"
28 / 0000	16.9	114.3	939	125	"
28 / 0600	17.3	115.0	943	120	"
28 / 1200	17.6	115.7	950	110	"
28 / 1800	17.9	116.5	955	110	"
29 / 0000	18.1	117.3	948	115	"
29 / 0600	18.3	118.0	952	110	"
29 / 1200	18.5	118.7	960	100	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
29 / 1800	18.9	119.6	968	90	"
30 / 0000	19.3	120.5	970	85	"
30 / 0600	19.8	121.5	973	80	"
30 / 1200	20.2	122.9	980	70	"
30 / 1800	20.6	124.2	984	65	"
31 / 0000	21.1	125.5	989	60	tropical storm
31 / 0600	21.5	126.7	993	55	"
31 / 1200	21.9	127.9	996	50	"
31 / 1800	22.2	129.3	996	50	"
01 / 0000	22.4	130.7	1000	45	"
01 / 0600	22.5	131.9	1000	45	"
01 / 1200	22.8	133.1	1001	45	"
01 / 1800	23.1	134.4	1002	45	"
02 / 0000	23.4	135.5	1003	45	"
02 / 0600	23.8	136.7	1003	45	"
02 / 1200	24.2	137.8	1006	40	"
02 / 1800	24.6	138.9	1008	35	low
03 / 0000	25.0	140.0	1009	35	"
03 / 0600	25.3	141.0	1009	35	"
03 / 1200	25.7	142.0	1009	30	"
03 / 1800	26.0	143.0	1010	30	"
04 / 0000	26.5	143.8	1010	25	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
04 / 0600	26.9	144.3	1010	25	"
04 / 1200	27.3	144.6	1011	25	"
04 / 1800	27.5	145.0	1012	20	"
05 / 0000	27.7	145.4	1013	20	"
05 / 0600	27.7	145.8	1013	20	"
05 / 1200	27.7	146.3	1013	20	"
05 / 1800	27.8	146.7	1014	20	"
06 / 0000	28.0	147.0	1015	15	"
06 / 0600					dissipated
27 / 0600	15.0	111.7	924	140	maximum winds and minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	168-Hour Outlook
Low (<40%)	54	174
Medium (40%-60%)	36	126
High (>60%)	12	78

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Genevieve, 24 July–2 August 2026. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**12.6**	**22.0**	**27.2**	**30.3**	**30.5**	**28.4**	**31.0**	**50.9**
OCD5	24.8	53.0	83.4	120.0	150.0	182.8	195.9	203.8
Forecasts	35	33	31	29	27	25	21	17
OFCL (2021-25)	21.0	31.9	41.7	50.4	60.8	73.3	99.6	124.1
OCD5 (2021-25)	35.9	72.5	113.0	154.2	196.4	239.0	325.0	405.8

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Genevieve, 24 July–2 August 2026. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	12.6	22.0	27.2	30.3	30.5	28.4	31.0	50.9
OCD5	24.8	53.0	83.4	120.0	150.0	182.8	195.9	203.8
GFSI	13.4	23.7	32.4	43.6	51.0	58.7	82.2	89.9
HWFI	15.4	23.6	28.8	**30.2**	34.2	42.7	69.0	104.9
HMNI	**12.5**	**20.9**	29.8	39.7	47.7	58.0	81.3	107.7
HFAI	14.2	**20.1**	**21.4**	33.0	46.4	65.8	111.4	135.0
HFBI	14.6	**19.3**	**21.9**	34.5	45.6	62.9	112.3	149.4
EMXI	12.8	22.5	32.9	45.7	56.9	67.5	88.8	117.4
NVGI	27.8	53.6	84.4	117.0	141.4	161.1	194.0	210.2
CMCI	17.0	28.1	34.7	39.0	42.4	43.8	45.2	80.4
CTCI	15.3	24.0	29.7	36.4	41.5	49.8	78.9	107.7
GDMI	**11.9**	**21.7**	29.0	33.7	33.7	33.0	32.8	53.1
TVCN	**11.3**	**17.2**	**23.6**	**30.0**	33.8	43.7	66.5	88.1
GFEX	**12.4**	**21.2**	31.5	42.4	49.7	57.5	79.0	95.3
HCCA	**11.5**	**21.5**	**26.5**	32.2	34.9	37.1	46.2	77.3
FSSE	**12.4**	**21.0**	32.0	42.8	51.2	59.5	86.9	115.0
AEMI	12.8	23.8	32.1	42.7	47.4	58.1	86.6	114.5
TABS	26.6	50.9	68.2	79.1	86.4	91.1	112.5	132.6
TABM	21.2	40.2	57.9	72.8	81.0	87.4	87.8	123.1
TABD	23.5	50.3	75.8	97.0	110.1	119.9	104.0	82.0
Forecasts	35	33	31	29	27	25	21	17

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Genevieve, 24 July–2 August 2026. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**5.3**	**8.5**	**10.0**	**10.3**	**8.5**	**5.8**	**5.0**	**5.3**
OCD5	7.8	12.6	15.5	16.3	15.7	14.7	12.9	8.6
Forecasts	35	33	31	29	27	25	21	17
OFCL (2021-25)	5.6	8.8	10.9	12.7	13.7	14.0	15.5	16.7
OCD5 (2021-25)	7.6	12.6	16.4	19.4	21.1	21.2	21.0	21.9

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Genevieve, 24 July–2 August 2026. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.3	8.5	10.0	10.3	8.5	5.8	5.0	5.3
OCD5	7.8	12.6	15.5	16.3	15.7	14.7	12.9	8.6
HWFI	8.9	11.0	13.3	14.6	13.0	11.4	9.3	6.1
HMNI	6.0	**8.1**	10.9	12.8	11.2	8.6	8.7	7.5
HFAI	7.9	10.5	11.9	13.1	13.1	11.9	8.4	7.0
HFBI	7.6	10.4	11.7	13.3	13.2	11.4	8.8	6.9
DSHP	6.9	9.5	10.4	12.5	13.6	12.4	10.9	10.1
LGEM	6.9	10.7	13.1	13.9	12.0	8.6	6.0	7.4
IVCN	6.4	8.9	10.5	11.7	10.9	8.8	7.1	**4.9**
CTCI	5.8	**8.2**	**9.9**	11.3	12.0	9.8	10.3	6.7
GFSI	6.9	11.1	14.3	16.7	18.1	16.2	10.6	**3.6**
EMXI	9.0	15.3	20.4	23.7	21.8	18.4	13.1	6.5
HCCA	6.6	10.2	10.1	10.7	10.7	7.4	9.6	15.2
FSSE	6.0	**7.6**	10.2	11.2	10.5	9.5	11.0	9.7
GDMI	7.6	10.4	12.7	13.8	13.9	12.8	12.4	13.4
Forecasts	35	33	31	29	27	25	21	17

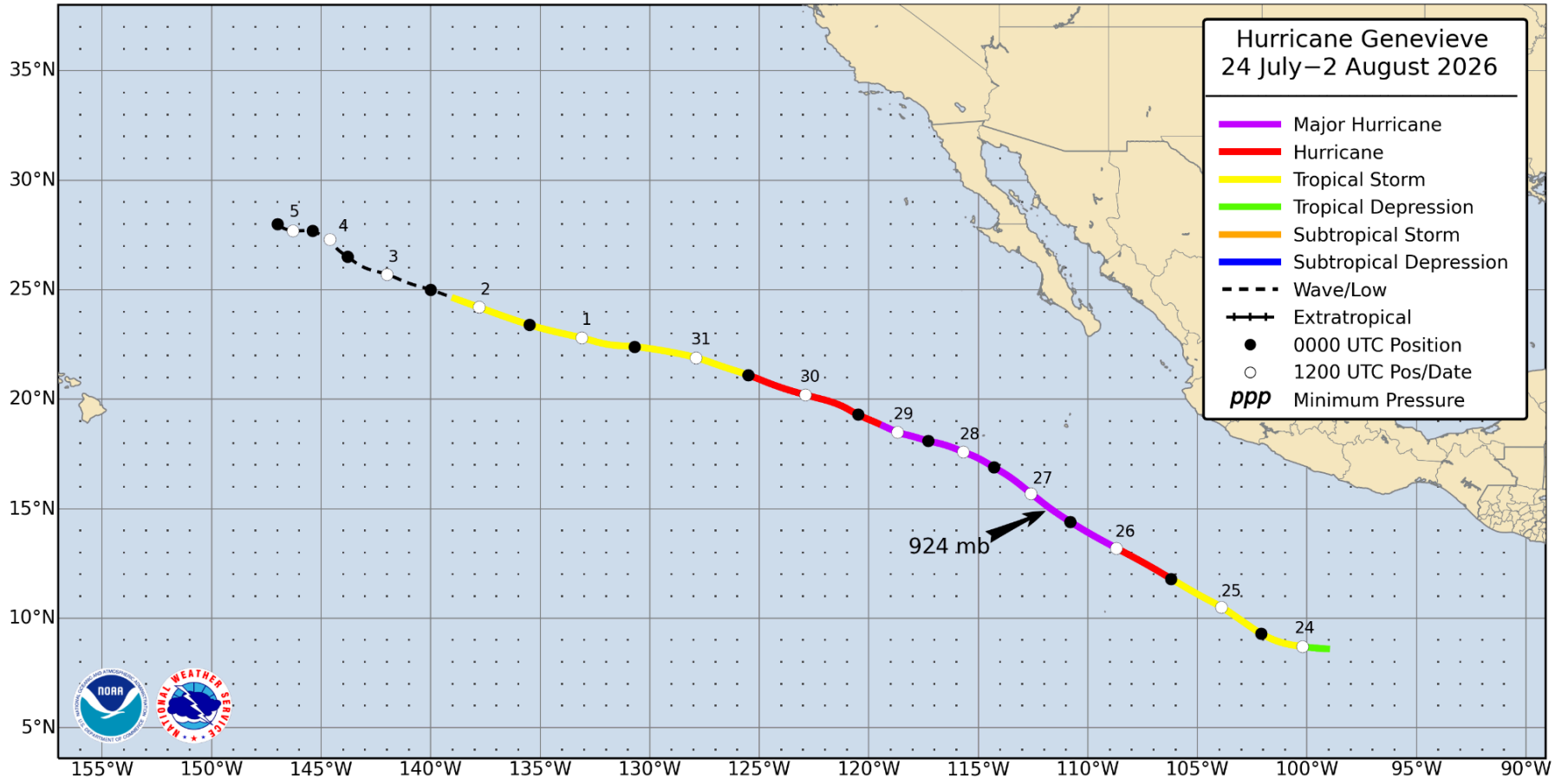


Figure 1. Best track positions for Hurricane Genevieve, 24 July–2 August 2026.

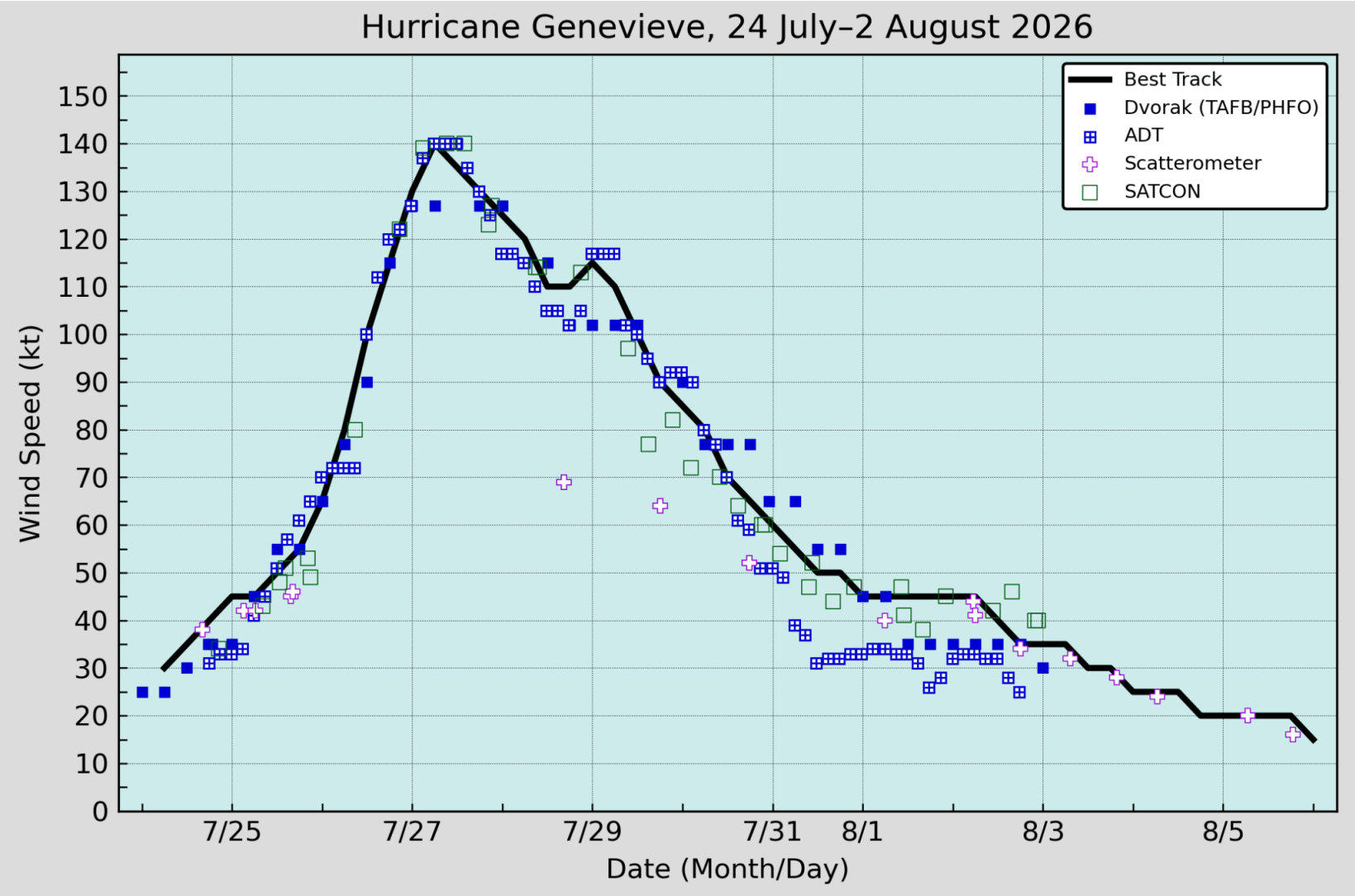


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Genevieve, 24 July–2 August 2026. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

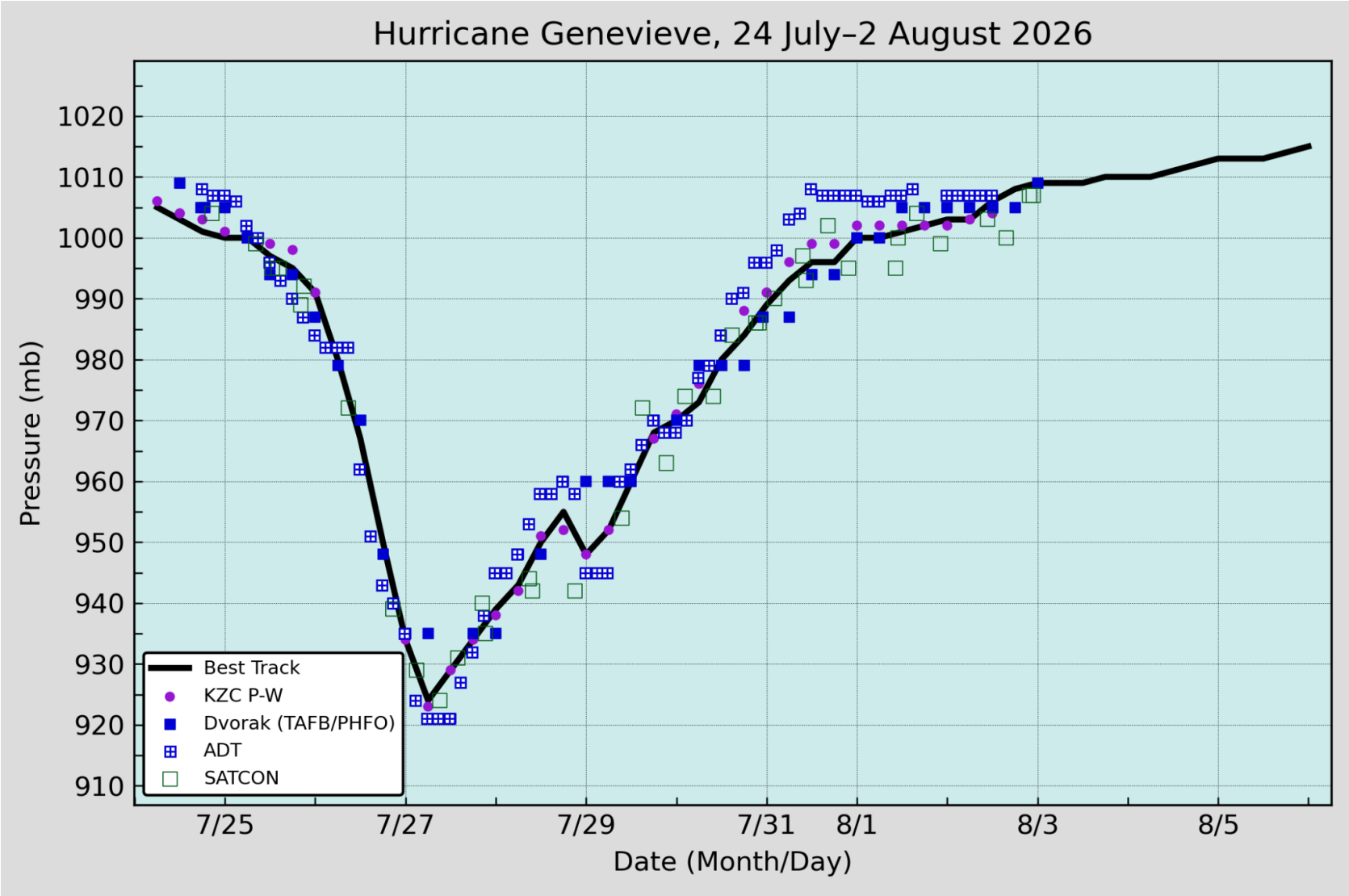


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Genevieve, 24 July–2 August 2026. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

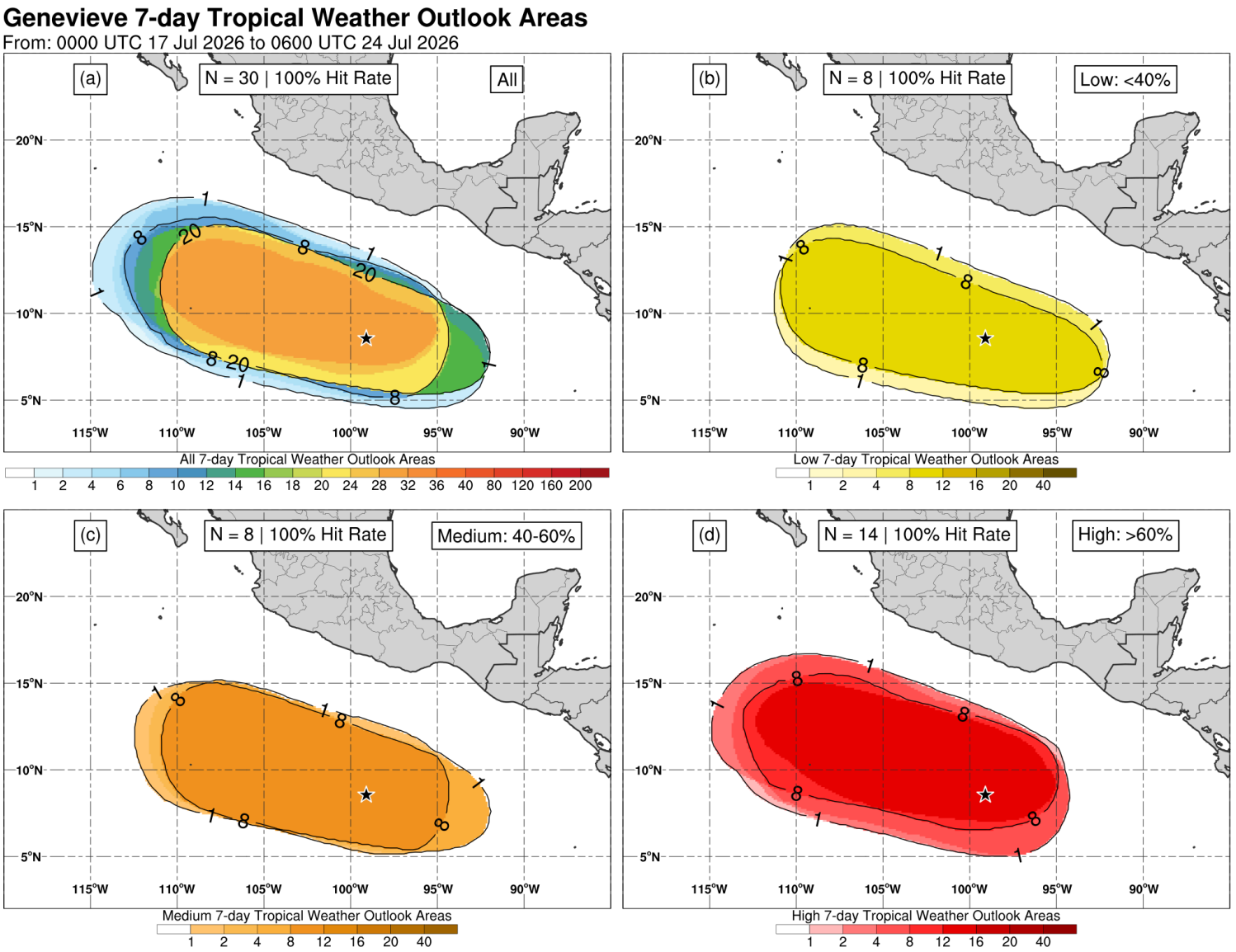


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Genevieve for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.

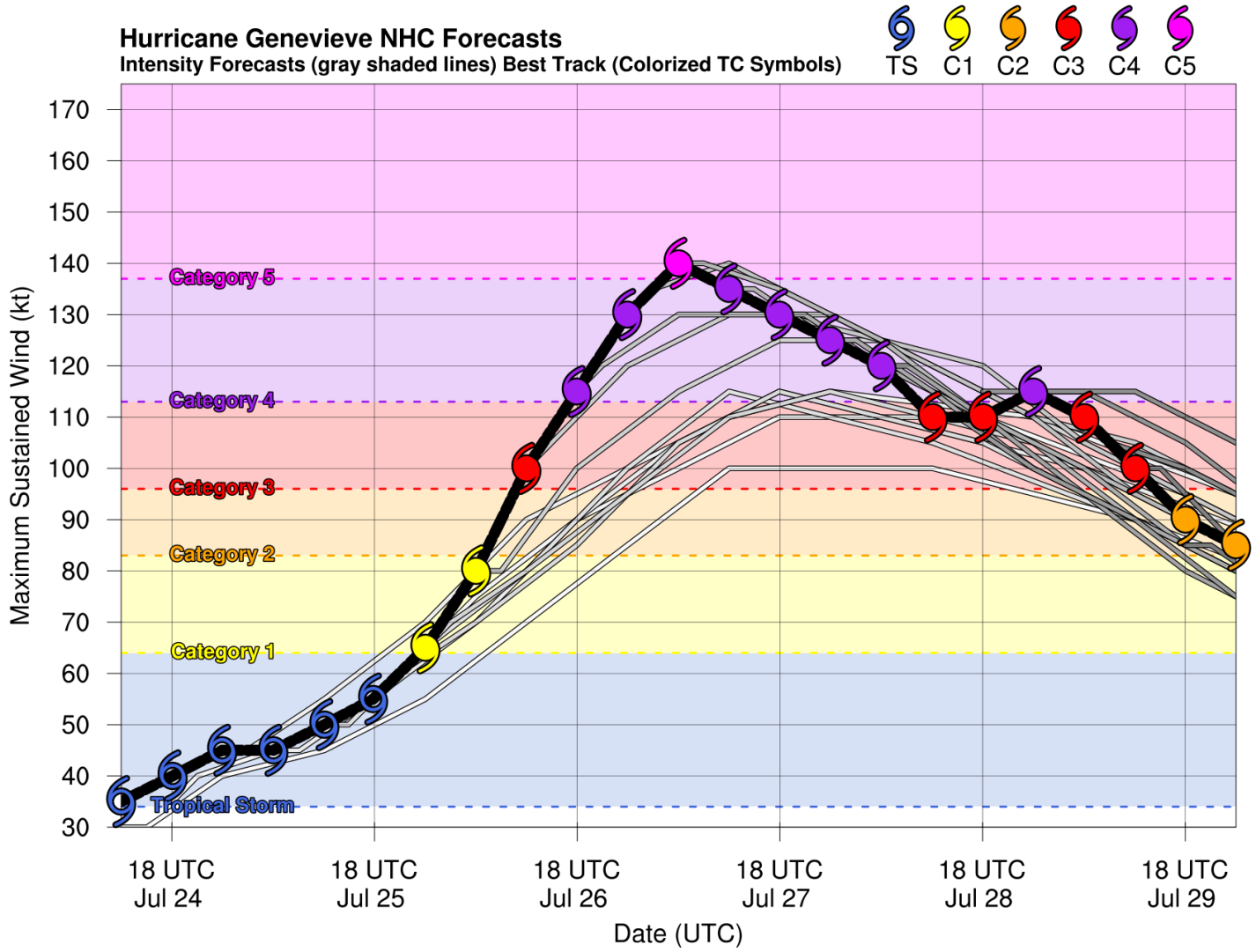


Figure 5. Official NHC intensity forecasts (OFCL) vs best track intensity for Hurricane Genevieve for the time during rapid intensification 24 –29 July 2026.